

FILE ALLOCATION TABLE

SPECTRAVIDEO USERS GROUP *WGTN*

NEWSLETTER NUMBER 6 NOVEMBER 1984

The next meeting is on MONDAY , DECEMBER 17 , at 7.30 in the STAFF COMMON ROOM, Floor D , Wellington Clinical School , Mein Street , Newtown. A committee meeting will be held at 6.30 .

There will be NO newsletter or meeting in January. The next meeting will be arranged for near the end of February. A newsletter will also be sent out that month. This is the last newsletter for 1984. On behalf of the club committee I would like to say thank you all for helping in a successful first few months for the club. I only hope that you all found the club worthwhile. Keep the newsletter articles coming in , we can't function without them , and have a Merry Xmas. See you all in 1985 !!

And a special word of thanks to Moonshine and Einstein computers in Wellington for their assistance in 1984 to the club. We look forward to continuing our close liason with both companies and other dealers in the vicinity.

Wellington SpectraVideo Users Group

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Club Committee

Don Stanley	ph 896379
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Dennis Timmons	ph 837200
Ross Wakelin	ph 374686
Mike Dalliessi	ph 848416
Darryl Alison	ph 85522 (h) or 88049 (w)

These are all home numbers , feel free to ring us about club matters (or problems/newsletter articles etc) , but give us a break after 9.00 Ok. (and during MAGNUM !!)

Note that each committee member is responsible for some area of the club , I look after the newsletter and general secretarial things , Colin looks after money and is also the MSX link , Dennis is responsible for the library , Ross is our national committee rep , Mike is keeping track of overseas users groups

and liasoning with them where necessary , and Darryl is our Kapiti person , feel free to ring him if you live in the Kapiti area.

Notes from meeting of November 19

We had a committee meeting in which we decided that at present the club should promote itself as a SpectraVideo club and not currently as an MSX club. While we intend to support SpectraVideo MSX we feel that at present we have not got the resources to cope with two really separate types of systems. We will be constantly monitoring MSX developments and reporting on them in newsletters but will be putting our major emphasis on SpectraVideo . The club will be known officially as the SpectraVideo Users Group (Wgtn) .

We also discussed problems with the printing of newsletters. We have had difficulties with getting the newsletter printed on time and are investigating ways of overcoming this. Meanwhile please accept our apologies for the late arrival of last months newsletter , it arrived back from the printers 4 days before the meeting and although I posted them the same day they did not arrive until after the meeting . Sorry , but it was right out of our hands.

We have decided on a policy of not making or allowing copies of any library material which is currently on sale in this country. This includes KIWIMSX magazine , if you want it -- subscribe to it.

A note of interest from the meeting itself concerns bugs in the Basic on the SV. Two points were raised ..

- 1 .. there is a bug of some sort in the A option of the DRAW command which occurs at times. It does not always occur , but for some reason A1 and A3 sometimes bomb out .
- 2 .. the circle command does not behave as the manuals say when joining the endpoint to the centre. The manuals claim that using a negative start or end point causes the start/end to be joined to the centre and the start/end point to be treated as positive. In fact the following applies ...

a) if the start point only is negative then the endpoint is joined to the radius but the start point is taken as zero

b) if the endpoint only is negative , or both are negative , then the endpoint is not joined to the centre and both points are considered positive.

ODDS AND ENDS ...

Bits and Bytes magazine want to publish SpectraVideo programs in their magazine . They pay for programs , so if you have programs , even small simple ones contact Darryl Alison ph 399510 for more details.

Members ... welcome to ... Graeme Dick , Ray Bush , Lawrence Hodson , R E Plummer , John and Judith Miller , Dan Evison , R Springer

Members Discounts ... Moonshine Computers in Lower Hutt and Wellington as well as Einsteins Computers are offering discounts on selected goods to members .. enquire at the shops for details. Use membership cards to verify membership when buying discounted goods from these stores. Note that the Einstein discount applies to their stores COUNTRY WIDE.

Contributions to this newsletter are always welcome. Preferably type them on A4 with a good quality ribbon. Anything from software/hardware reviews through to notes on Basic , CP/M , machine code or whatever are acceptable . Send to Don Stanley. Also those of you doing unusual things with your SV eg interfacing with other machinery , lets have a short article on what you are doing and any hints for others considering doing similar things. Also most programs are initially written for cassette. If you convert them to disk , let us know and send details . If you use a computer printer to send articles , please use emphasized print if possible. Note that the deadline for the newsletter going to the printers is the 30/31 of each month.

Book review -- Peeks & Pokes For the Spectravideo

This latest addition to the SV book list is written by Deane Whitmore from the Taranaki users group. It is packed full of the sorts of gems that are hidden in the depths of the SV.

As the title suggests a lot of the book is devoted to using some of the memory locations which contain system values. How Deane found some of these I'll never work out , who on earth would think of poking 7B into location 64003 and then holding down the backspace key -- Deane would !!

The book is packed full of examples including many small illustrative programs and some bigger programs. An excellent look into the video ram , including where you'll find your sprites is provided . While the book doesn't look large don't judge it on

size , this is a must for anyone wanting to find out more about the SV's innards -- definately recommended.

SPECTRAVIDEO REVIEW 'SECTOR ALPHA' by Alex Bridger.

Here is a fighter pilot simulator game that is different to the usual (mind you so is the price). The setting is three different mountainous landscapes from a planet somewhat reminiscent of science fiction film settings of a different solar system.

You are required to defend your planet against invading alien Tie fighters that attack you in the three sectors regardless of which sector your craft happens to be in. You have three radar screens that give you the overall view of the mounting attacks indicating which Sector will next require your attention. You launch missiles at the incoming fighters which in turn launch missiles at you. These enemy missiles very realistically grow as they get closer and can only be evaded provided you act soon enough before they smash into your cockpit and terminate one of your five lives. The most fascinating aspect of this game is the realistic shifting of the scenery in response to your aircraft joystick movement. (I have never flown a Skyhawk to compare the feeling but its not too dissimilar to light aircraft flying.)

A beginner pilot feels considerable overload of information (the scoring is tricky) but with half an hour of dual instruction and 2 hours solo the game becomes straightforward but still fun. (After a week we were in the six figure class.)

The cost of around \$130 for this ROM cartridge is the main drawback for such an enjoyable game , however sharing one between 5 people who live reasonably close can reduce this to a reasonable cost. It runs on the SV318 or 328 with no attachments. Beware of buying this sort of game in Australia ,in Sydney it was priced between NZ\$105 and \$170 before paying up to 80% customs duty!

GAMES SCORES

SASA	highest score to date	45120	(Michael Wright)
TURBOAT		68390	(Garth Thornton*)
TETRA H		256750	(Colin Chin)
NINJA		53400	"
OMAC		25700	(Don S+)
SPECTRON		8490	(Colin Chin)
TELEBUNNY		87810	"
SUPER CROSS FORCE		748600	(Deirdre Chin)
SECTOR ALPHA		147000	(Barbara Bridger)
KUNG FU		15835	(Angela Chin)

Lets hear about your high scores , BE HONEST , maybe we'll have a games freaks king of the Spectra competition one night to check it all out !!!!

Note that a * in the scores means the game was started in ARCADE level , a + means normal level or equivalent and neither means practice level or unknown.

Please note that I am no longer employed by the Hospital Epidemiology Unit . I can be contacted by phone at the BNZ , phone 746799 and ask for Management Information Services and me , but please send all mail to P.O Box 7057 .

Don Stanley

The File Allocation Table -- and Bad Allocation Tables

I have been pulling the Disk operating system to bits to try and get some idea why the bad allocation table problem occurs , here are the results and a few ideas.

The file allocation table is essentially a map of how a file is stored on the disk. When a file is stored there is no reason why it should be stored as most of us would store a file ie all together , instead it is likely to be stored in several different parts of the disk.

Lets look at the file directory first. This occurs on track 14h = 20 decimal on every disk. No files can be written on this track as it is reserved solely for storing information about the files on the disk. Each file has 16 bytes of information on this track , these are as follows ..

bytes 0-8	filename
9	file type
10	pointer into allocation table
11-15	reserved

If a file is deleted, byte zero is given the value zero. Byte 9 was discussed in newsletter number 2. The byte of interest here is byte 10.

Also on track 20 is the file allocation table. It occurs in sectors 15,16,17. As previously explained it is a map of the disk, the entry point to the map for a given file is byte 10 in the directory table. Note that sectors 15,16 and 17 are identical.

Lets look at a typical allocation table, all 255 bytes worth. The following table is from a disk with 1 file on it called format.

```
FE FE FE FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF C5
FE FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
0 0 0 FF 0 E FE FE FE FF FF FF FF FF FF FF C4 C5 C7 C5 C4
C4 C2 D1 CF 11 C5 FE C4 C2 18 C8 C2 1B 1C CC 1E C7 C2 FF FF
FF FF FF FF FF FF 0 C1 F1 F5 CD 68 DA CD 8C DB F1 C1 D1 E1
C9 4F 21 34 FE 7E A7 C2 18 DB 79 FE 20 38 21 2A 3 FA FE 7F
CA 32 DC CD 57 DD CD A3 DB C0 AF CD 56 1 26 1 CD BD DB C0
CD C5 DB 2E 1 C3 E4 DB 21 B0 DA E C 23 23 A7 D F8 BE 23
20 F7 4E 23 46 2A 3 FA CD B0 DA AF C9 C5 C9 7 47 1 8 AA
DB 9 CD DB A 8B DA B DB DB C 6A DC D DD DB 1B 12 DB 1C
B7 DB 1D AA DB 1E B3 DB 1F BD DB 6A 6A DC 45 6A DC 4B 3D DC
4A 6D DC 6C 3B DC 4C B DC 4D E1 DB 59 F DB 41 B3 DB 42 BD
DB 43 A3 DB 44 B1 DB 4B DB 4B 70 B6 DB 71 97
```

Firstly note the first 3 bytes are all hex FE. Essentially each byte here points to a track. The position of the byte is the track number. Note that the SV manuals use the word cluster to refer to a track -- a cluster is a track, NOT half a track as some manuals would have us believe. Tracks 0,1 and 2 (the first 3) are the system tracks and cannot be altered, except with DSK0%. The entry in the file allocation table (fat) is always 254 for these tracks. (Incidentally, DSKF(drive num) tells us the number of unused tracks, which is the maximum number of files you can put on the disk before filling it).

The entrys equal to 255 are a flag that the cluster is free to be written on. Any of these may be entry points from the directory byte 10, or may be used as part of the disk map.

There is a further FE at track 20. This is the directory track itself, the FE tells the DOS not write on this track.

Note that the fat sector is 255 bytes long. But only the first 40 bytes are used. There are 215 wasted bytes in the fat for each sector. This is due to the smallest unit of storage being a track, consequently only 40 bytes are needed to completely define the map. The implication for storage is that only 40-1-3 = 36 files can fit on a disk at most. But the sectors reserved for the directory number 13 at 255 bytes each ie 3315 bytes. Only 36*16=576 bytes can be utilised, the SV DOS

wastes $215 \times 3 + 3315 - 576 = 3384$ bytes on the directory track alone , unfortunately still more are wasted due to the smallest storage unit being a track. (It means that a file 1 byte long is considered logically equivalent , from the DOS point of view , to one a whole track long)

Back to file allocation tables. There is one odd byte in the first 40 of the above fat , the C5 at position $13h=19$ decimal. But think back to the file directory. For this file , the directory bytes are ...

```
66 6f 72 6d 61 74 20 20 20 80 13 ...
f o r m a t
```

The entry point is $13h$ (the byte at position 11) which is our C5. In fact $C5 = 192+5 = 197$. Using page 24 of the disk basic users guide , note that the last entry in the map for a file is equal to $192 +$ the number of sectors used from this track. Hence the entry point , C5 is also the last part of the map for this file. Thus the file physically occupies 5 of the 612 available sectors or 0.82% of the available disk space. But since the track is the smallest unit of allocation , the file really fills up $1/36 = 2.78\%$ of the disk storage .

Lets look at a more complex example ... which just happens to have a bad allocation table ...

The allocation table is as follows , with entry tracks 7,16, 18,13 and 15 (all hex) .

```
FE FE FE FF FF C6 5 6 FF FF FF FF FF FF FF FF FF FF
FE 11 17 CF 19 1A 1B 1C 1D 1E 1F C7 FF FF FF FF FF FF FF
0 0 0 FF 0 E FE FE FE FF FF FF FF FF C4 C5 C7 C5 C4
C4 C2 D1 CF 11 C5 FE C4 C2 18 CB C2 1B 1C CC 1E C7 C2 FF FF
FF FF FF FF FF FF 0 C1 F1 F5 CD 68 DA CD 8C DB F1 C1 D1 E1
C9 4F 21 34 FE 7E A7 C2 18 DB 79 FE 20 38 21 2A 3 FA FE 7F
CA 32 DC CD 57 DD CD A3 DB C0 AF CD 56 1 26 1 CD BD DB C0
CD C5 DB 2E 1 C3 E4 DB 21 B0 DA E C 23 23 A7 D FB BE 23
20 F7 4E 23 46 2A 3 FA CD B0 DA AF C9 C5 C9 7 47 1 8 AA
DB 9 CD DB A BB DA B DB DB C 6A DC D DD DB 1B 12 DB 1C
B7 DB 1D AA DB 1E B3 DB 1F BD DB 6A 6A DC 45 6A DC 4B 3D DC
4A 6D DC 6C 3B DC 4C B DC 4D E1 DB 59 F DB 41 B3 DB 42 BD
DB 43 A3 DB 44 B1 DB 4B DB DB 70 B6 DB 71 87
```

Calling the files A to E for simplicity , we get the following.

FILE A -- starts on track 7. The allocation table byte at position 7 is 6 , less than 192. Hence we look for the byte at position 6 , this is 5. We look at the byte in position 5 this is $C6 = 192+6$. This is the last track. The file uses tracks 5 , 6 and 7 ie as far as the DOS is concerned $3/36 = 8.33\%$ of disk space. Physically it uses $(17+17+6)/612 = 6.54\%$ of the disk space.

FILE B -- starts on track 16h = 22 decimal. The byte is a 17 indicating that track 17h is also used. The byte at the track 17 position is CF = 192+15. The file thus uses 2 tracks, numbered 16 and 17.

FILE C -- starts on track 18h = 24 and stretches across the next 8 tracks.

FILE D -- starts on track 13h = 19 decimal. The byte is an FF. But an FF is reserved to tell the DOS the track is free. Here is a source of the file allocation table error.

FILE E -- starts on track 15h = 21 with the byte being 11h = 17. Track 17 is an FF ie another allocation table error.

If one thing becomes obvious from the above it is that fat errors cannot be easily recovered from. FILE D is actually 5 clusters long. But all these unallocated clusters are set to FF, finding the relevant ones in the right order would be horrendous. guess it could be done slowly and carefully with DSK0\$ and DSKI\$.

Note that the problem affects the files statement which actually uses the directory sectors and the allocation table. This is why a files will often bomb after printing the filename but prior to printing the number of used tracks.

What causes the problem to occur? Note that it affects the files on either side of the directory track. I can't see how this could cause difficulties UNLESS the heads are poorly designed and can sometimes fall out of alignment near the reference track ie the directory track. I have another disk which has the same two tracks corrupted. The possibility of a hardware fault in the drives is not to be excluded. Its harder to believe that software could cause such a sporadic fault.

Investigating the allocation table has shown another interesting fact about the SV DOS. Many systems, when a file is deleted, change only a byte in the directory, the disk map is not touched. Thus accidental deletions can be recovered by changing just one byte, provided of course that it is done before that part of the disk is reused. Not so the SV. A deletion sets the allocation map for the file to all FF's but doesn't alter the entry track byte in the directory. This is the reverse to many systems and is a virtual guarantee that a KILL is a real kill even though the file still physically exists.

Running the following program on a disk with a bad allocation table will pinpoint the corrupted file(s). Deleting them should solve the problem, but of course ... byebye files.


```

10 SCREEN,0:WIDTH39
20 CLEAR 1000
30 COLOR 1,15
35 DIM DF$(40)
40 RV$=CHR$(27)+"p" : NV$=CHR$(27)+"q"
41 FOR I=4 TO 40:DF$(I)="Unallocated":NEXT
42 FOR I=1 TO 3:DF$(I)="System Use":NEXT
43 DF$(21)="Directory"
50 FOR S=1 TO 13
60   A$ = DSKI$(1,20,S)
70   FOR I=1 TO LEN(A$) STEP 16
80     IF ASC(MID$(A$,I,1)) = 0 THEN 270
90     IF ASC(MID$(A$,I,1)) = &HFF THEN 290
100    B$ = MID$(A$,I,9)
110    EP$ = HEX$(ASC(MID$(A$,I+10,1)))
120    CLS
130    Y=8:TR=1
140    LOCATE 4,1:PRINT "FILE ";B$;
150    LOCATE 4,3:PRINT "Allocation Table Entry Track ";EP$;"h"
160    LOCATE 4,5:PRINT "Allocation Table Map "
170    AT$=DSKI$(1,20,15): AT$ = LEFT$(AT$,40)
175    IF DF$(VAL("&h"+EP$)+1)="Unallocated"THEN DF$(VAL("&h"
*   +EP$)+1)=B$ ELSE DF$(VAL("&h"+EP$)+1)=B$+CHR$(215)
180    PO$=HEX$(ASC(MID$(AT$,VAL("&h"+EP$)+1,1)))
190    IF VAL("&h"+PO$) = &HFF THEN LOCATE 4,20:PRINTRV$"Corrupt
*   Allocation Table"NV$: DF$(VAL("&h"+EP$)+1)=RV$+"Corrupted"+
*   NV$:GOTO 250
200    IF VAL("&h"+PO$) >=&H00 THEN LOCATE 4,20:PRINT"End of Map
*   -- Tracks Used =" ;TR:GOTO 250
210    LOCATE4,Y : PRINT "---> Next Track ";PO$ : EP$=PO$
220    Y=Y+1:TR=TR+1:IF DF$(VAL("&h"+PO$)+1)="Unallocated"THEN
*   DF$(VAL("&h"+PO$)+1)=B$ ELSE DF$(VAL("&h"+PO$)+1)=B$+
*   CHR$(215)
230    IF Y=20 THEN Y=8
240    GOTO 180
250    LOCATE 5,21:PRINT"Press a key to continue"
260    IF INKEY$="" THEN 260
270    NEXT
280 NEXT
290 LOCATE5,21:PRINT SPC(30):LOCATE5,21:PRINT"No more files --
*   Summary Follows"
300 FOR I=1 TO 300:NEXT
310 CLS
320 Y=2:LOCATE3,0:PRINT"Track":LOCATE23,0:PRINT"Track"
330 FOR I=1 TO 40
340   LOCATE-3*(1<=20)-23*(I>20),Y:PRINT I-1;" ";DF$(I)
350   Y=Y+1 : IF Y=22 THEN Y=2
360 NEXT
370 LOCATE0,23:PRINTRV$"Press any key to end"NV$;:IF INKEY$=""
*   GOTO370 ELSE LOCATE0,0:END

```

In the program output a downwards arrow indicates a track which has been doubly allocated. This is a form of fat error, in fact both files will be the same as they have the same entry track.

In the program listing a * in the line number field indicates a line which is a continuation of the preceding line. In most cases the blanks on these lines can be ignored.

I have just had the opportunity to talk to Doug Greig from Auckland. Doug suggests that if you get a bad allocation table error you immediately type CLOSE and then FILES. If the error occurs again try taking the disk out of the drive and redoing the above procedure. It works sometimes, but not every time I'm afraid.

If you're getting annoyed with the disk drives not turning off when you load a file try this ...OUT 52,0 .

or alternatively patch the DOS using the following program to alter the 2 bytes which control the spin.

```
10 '  
20 ' patches the DOS to prevent the disk spinning all the time  
30 '  
40 CLEAR2000  
50 A$=DSKI$(1,0,12)  
60 MID$(A$,20,2)=CHR$(50)+CHR$(0)  
70 FIELD #0,128 AS B$,127 AS D$  
80 LSET B$=LEFT$(A$,128)  
90 LSET D$=RIGHT$(A$,127)  
100 DSKO$ 1,0,12
```

This modification will switch the disk off NEARLY instantly whenever you carry out a disk operation without needing the OUT statement. The patch is carried out on a sysgened disk basic disk only.

Notes on Function Keys

The function key definitions are stored in ram starting at the location &hfale . By using the command KEYn," text " these can be altered within the 15 character field allocated.

There is another means of altering the keys contents, using the basic POKE command. This is rather interesting, you can poke up to 39 characters for each key. The key is terminated by a zero byte. Poking a return does not terminate the key definition, try the following short program to see what happens when a return is not followed by a null character for key 1 .

```
10 for i=&hfale to &hfale +35 step 2  
20 poke i,64 : poke i+1,13  
30 next  
40 poke &hfale+36,0
```

Now press F1 , you should get a lot of syntax errors , as the function key is actually 35 bytes long with returns inserted all through it .

Actually , saying you can poke up to 39 bytes per key is a bit misleading. Each key can reference up to 39 bytes , but they will overlap , as key 2 always starts at position õ17 (there are 15 characters for each key , then 1 null (ie zero byte) for each key to terminate) and so on .

One use I have found for these super long F keys is to set up a program which pokes load instructions for several files on a disk into them. Then at the press of a single key ... you can load and run a file. You can only use every second key though , because of the overlap.

New public domain software

Elsewhere in this newsletter is an article from Peter Thakurdas about transferring files between Basic and CPM. Both the programs listed , cpm.it and basic.it , are available as public domain programs under disk basic.

We have several programs from Deane Whitmore now in the library . These include subhunt and galaxy , two games with very nice graphics , as well as utility programs and a couple of graphics screen saves of 3-d plots.

The public domain software we have now fills 3 disks. If you want copies send me disks formatted for disk basic. I will copy only whole disks, not individual files. Tapes can be done also , but are likely to take longer to do. I would prefer not to do tapes , so if you know someone with a disk , maybe you could get copies from them.

Another SV book has hit the shops. This has been written in Australia and is called Games for the Spectravideo. Available from Whitcoulls at least .

I have often wondered about how TURBOA uses the shift key. If you haven't played it , you need only press the shift key to make your boat shoot automatically. The MSX users guide from Microsoft as the answer as to how the SV does this , and also shows how to trap pressing of the CTRL , LGRP and RGRP keys.

So try this ...

```

10 y = peek(&hfd86)
20 x= y and &b00001111
30 if x=15 then 10
40 else if x=14 then print "SHIFT KEY":goto 10
50 else if x=13 then print "CTRL KEY":goto 10
60 else if x=11 then print "LEFT GRPH":goto 10
70 else if x=07 then print "RIGHT GRP":goto 10
80 goto 10

```

Memory location &hfd86 monitors the status of the 4 keys. When no key is pressed the status is 15 or a binary 1111 as above .

.. More goodies from Taranaki

```
for x=&hfe79 to &hff11 step 3 : poke x , &hc9 : next
```

will switch out the functions to the disk operating system.

POKE 65079,1 will cause the cursor to switch to the insert mode cursor .. (does anyone know if the cursor can have the shape changed in any column mode ??????)

and a query from Deane in Taranaki, does anyone know how to list protect a program ie switch off the list function. If so how about sharing it round.

THE CAPS LOCK KEY -----

A poke has recently been discovered that can toggle the CAPS LOCK key on and off. However, this does not change the light in the key. By using a ROM routine, a much more effective way can be used.

```
10 DEFUSRO=16359:A=USRO(0)
```

This changes the CAPS LOCK key from on to off, or from off to on. To check whether the key is on or off, the following should be used.

```
10 X=INP(144) AND 32
```

X=32 if the computer is in capitals, and anything else if it is in lower case.

From ELLIOTT YOUNG

The following outlines methods for transferring files CPM <--> BASIC.
Also provided are two BASIC programs which are needed, "cpm.it" and "basic.it".
"cpm.it" and "basic.it" have been made public domain utilities for use on Spectravideo 328 disk systems.

A. BASIC --> CP/M (Maximum file size is just under 32K)

1. Have the ascii file you want to transfer and the BASIC program "cpm.it" on a BASIC disk. (Note: To get an ascii version of a program on disk, use save "1:prog.txt",a).
2. Load and run "cpm.it" (eg. Type: load "1:cpm.it",r).
3. Give the name of the file at "Which file ?" prompt (eg. prog.txt).
4. Wait while the file is displayed as it is transferred.
5. When file transfer is complete, make a note of the instruction message (eg. SAVE 5 prog.txt).
6. Replace BASIC disk with CP/M disk, which has CP/M system and enough disk space. (In this example: at least 2K (= int(5/4) + 1).
7. Press any key to boot up CP/M disk.
8. FIRST thing after CP/M boot up, type (as instructed in (5) above) : (Eg. SAVE 5 prog.txt).
9. To check transfer worked, type: type prog.txt

B. CP/M --> BASIC (Maximum file size is 16K)

1. If you haven't already got G8000.COM (modified version of REBOOT.COM) then use DDT to patch REBOOT.COM (so that JP 0000 replaces JP 8000) and save it as G8000.COM. To do this, you'll need REBOOT.COM and DDT.COM (provided with CP/M), after CP/M boot up, type:

```
DDT REBOOT.COM
S10D
00
.
GO          (or CTRL-C)
SAVE 1 G8000.COM
```

2. Have DDT.COM (or equivalent), G8000.COM and file (to be transferred) on CP/M disk.
(Note: If file is a .HEX file, rename it to be say a .HXX file,
Eg. Type: REN NEWNAME.HXX=OLDNAME.HEX).
3. At the CP/M "A>" prompt, type: G8000
(This sets up the reboot-to-BASIC facility)
4. Type DDT followed by filename (Eg. DDT FILE.TXT)
(This gets the contents of the file into memory, ready for transfer)
5. Note number under "NEXT" (Eg. 1A00) once "in DDT" .
6. Replace CP/M disk with BASIC disk, which has DISK BASIC system, the BASIC program "basic.it" and enough room for the file being transferred
7. At the DDT "->" prompt, type: G8000
8. After BASIC disk boots up, load and run "basic.it"
(Eg. load "1:basic.it",r)
9. Give filename (one that doesn't already exist on disk)
at "New filename ?" prompt. (Eg. FILE.TXT)
10. Give number that was displayed under "NEXT" in DDT (noted earlier).
(Eg. 1A00).
11. Wait while file is displayed as it is written to disk.

```
5 REM: This program will transfer ascii files from BASIC to CP/M disks
6 REM: Author: Peter Thakurdas (1-NOV-1984)
7 REM: Public Domain Software for SV328 disk system
10 GOSUB 5000 : REM Initialise Machine Code Routine
20 GOSUB 1000 : REM do the transfer
30 PRINT "Swap BASIC disk for CP/M disk."
40 PRINT "Right after CP/M disk boots up,"
50 PRINT "Type:": PRINT: PRINT
60 PRINT "SAVE ";INT(N/256)+1;" ";F$;" <ENTER>":PRINT:PRINT
70 PRINT "Type any key to continue.":PRINT"Make sure CP/M disk is mounted."
80 A$=INPUT$(1)
90 DEF USR=0:PRINTUSR(0)
99 END
1000 INPUT"which file ";F$
1010 N=0
1020 OPEN"1:"+F$FORINPUTAS#1
1030 IF EOF(1)THENGOTO1080
1040 A$=INPUT$(1,#1): PRINT A$;: GOSUB 2000
1050 N=N+1
1060 IF EOF(1)THENGOTO1080
1070 GOTO1040
1080 CLOSE#1: A$=CHR$(26): GOSUB 2000
1090 PRINT,,, N;" bytes"
1100 PRINT CHR$(7)
1200 RETURN
2000 REM write char to RAM
2010 POKE &H87ED,ASC(A$)
2020 X=USR(0)
2030 RETURN
5000 REM gets data from BASIC to RAM beside ROM for transfer to CP/M
5010 J=&H8700 : REM Where the machine code resides
5020 READ A$
5030 IF A$="END" THEN GOTO 5150
5040 POKE J,VAL("&H"+A$) : J=J+1 : GOTO 5020
5050 REM Machine Code (hex) follows
5060 DATA f3,3e,0f,d3,88,db,90,e6,fd,d3,8c:REM ROM disable
5070 DATA ED,5B,EE,87 :REM LD DE,(087EEH) - destination
5080 DATA 3A,ED,87 :REM LD A,(087EDH) - source
5090 DATA 12 :REM LD (DE),A - transfer
5100 DATA 13 :REM INC DE - update destination
5110 DATA ED,53,EE,87 :REM LD (087EEH),DE - save new destination
5120 DATA 3e,0f,d3,88,db,90,f6,02,d3,8c,fb : REM ROM enable
5130 DATA c9 : REM RETURN
5140 DATA END:REM END machine code
5150 POKE &H87EE,0 : POKE &H87EF,1 : REM Corresponds to 0100H.
5160 DEF USR=&H8700
5170 RETURN
```

"basic.it" listing

```
-----
5 REM: This program will transfer data from CP/M (memory) to BASIC disk
6 REM: Author: Peter Thakurdas (1-NOV-1984)
7 REM: Public Domain Software for SV328 disk system
10 GOSUB 5000 : REM Initialise Machine Code Routine and transfer 16K from 0100H
20 PRINT "Up to 16K of data has been transferred"
30 PRINT "from CP/M (from 0100h onwards) to"
40 PRINT "RAM starting at 9000h."
50 PRINT "If it's not just ascii data, then"
52 PRINT "    CTRL-STOP to quit and use BSAVE"
54 PRINT "    Eg. BSAVE <file>,&h9000,&ha000"
60 PRINT:PRINT:PRINT "If it IS just ASCII data, then"
62 PRINT:PRINT "    press any key to continue..."
70 A$=INPUT$(1)
90 GOSUB 1000 : REM write to disk
99 END
1000 INPUT "New Filename ";F$
1010 INPUT "Number under 'NEXT' in DDT ";N$
1020 OPEN "1:"+F$FOROUTPUTAS#1
1040 FOR I=&H9000 TO ( &H9000 + VAL("&H"+N$) - &H100)
1050 A$=CHR$(PEEK(I)): PRINT #1,A$;
1060 PRINT A$;
1070 NEXT I
1080 CLOSE#1
1090 PRINT,,, VAL("&H"+N$) - &H100;" bytes"
1100 PRINT CHR$(7)
1200 RETURN
5000 REM gets data from RAM beside ROM for transfer to BASIC disk
5010 J=&H8700 : REM Where the machine code resides
5020 READ A$
5030 IF A$="END" THEN GOTO 5160
5040 POKE J,VAL("&H"+A$) : J=J+1 : GOTO 5020
5050 REM Machine Code (hex) follows
5060 DATA f3,3e,0f,d3,88,db,90,e6,fd,d3,8c:REM ROM disable
5070 DATA 11,00,90:REM LD DE,09000H - destination
5080 DATA 21,00,01:REM LD HL,00100H - source
5090 DATA 01,00,40:REM LD BC,04000H - counter
5100 DATA ED,B0:REM LDIR - transfer data
5120 DATA 3e,0f,d3,88,db,90,f6,02,d3,8c,fb:REM ROM enable
5130 DATA c9:REM RETURN
5140 DATA END:REM END machine code
5160 DEF USR=&H8700 : X=USR(0)
5170 RETURN
```

PROGRAMS from the RADIO

PROGRAMS from the RADIO

Every month starting from the 2nd February 1985, about 3 minutes of Public Domain computer programs will be transmitted by Radio Station 2 Y B (Access Radio).

What you do is to record off the radio onto cassette and then the tape is used in the normal way programs are loaded. CLOAD <enter>

Mark the Calendar now and take part in a Historical first for SPECTRAVIDEO and even for radio in New Zealand.

Time 9am exactly Saturday the 2nd February 1985 2YB 783 kHz.

Please note the station starts its regular broadcast for the day at 9am so that it would be best to either tune in the day before or on the day at about 8.30am when test signals and a test record are transmitted. Check the NZ LISTENER.

6845 CRT Controller Programmable Registers - From Philip Plare

No.	Register Name/Function	Read (R) Write (W)	Bits	Range-Units
0(00H)	Horizontal Total	W	8	1-256(0-FFH) CLK's
1(01H)	Characters/Row	W	8	1-256(0-FFH) CLK's
2(02H)	HSYNC Position	W	8	1-256(0-FFH) CLK's
3(03H)	HSYNC Width	W	4	1-16(0-FH) CLK's
4(04H)	Vertical Total	W	7	1-128(0-7FH) Char Rows
5(05H)	VSYNC Adjust	W	5	1-32(0-1FH) Scan Lines
6(06H)	Character Rows/Frame	W	7	1-128(0-7FH) Char Rows
7(07H)	VSYNC Position	W	7	1-128(0-7FH) Char Rows
8(08H)	Interlace Mode	W	2	0-3
9(09H)	Scan Lines/Row	W	5	1-32(0-1FH) Scan Lines
10(0AH)	Cursor Start Scan Line	W	7*	1-32(0-1FH) Scan Lines
11(0BH)	Cursor Stop Scan Line	W	5	1-32(0-1FH) CLK's
12(0CH)	(MSB) Start Address	W	6	
13(0DH)	(LSB) (Top of Page)	W	8	1-16,384(0000-4FFFFH)
14(0EH)	(MSB) Cursor	R/W	6	
15(0FH)	(LSB) Position	R/W	8	0-16,384(0000-4FFFFH)
16(10H)	(MSB) Light Pen	R	6	
17(11H)	(LSB) Position	R	8	0-16,384(0000-4FFFFH)

* Two bits used to specify cursor blink characteristics

To select a register, send the register No. out port &H50
Then send the data out port &H51.

The cursor blink characteristics are set by bits 6 and 5 of register 10.

BIT	6	5	Characteristic
	0	0	No blink
	0	1	No cursor
	1	0	Normal blink
	1	1	Half speed blink

To set the cursor start line send 0-7 to register 10, remembering to maintain the blink bits.

BASIC resets the cursor start line and blink rate when a cursor control key, tab key or enter key is used. INS sets the start line to 4 then 0 if pressed twice.

To set the cursor end line send 0-7 to register 11, remembering that the end line must be below or level with the start line to display a cursor.

If the start and end line are the same then a single line cursor is shown.

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3D FUNCTION PLOTTER

From HARRY PATERSON

```
10 COLOR 9,1,1:SCREEN 1:CLS
20 DEF FNA(A)=(SIN(X/3)*2)^3+(SIN(Y/3)*2)^3
30 FOR Y=0 TO 8 STEP.25:FOR X=0 TO 8 STEP.25
40 PSET(20*(X+Y),(Y-X+2+FNA(A))*6):NEXT X
50 LINE-(20*(Y+8),(Y-X+2)*6):NEXT Y
60 FOR X=0 TO 8 STEP.25:PSET(16*X,-6*X+12)
70 FOR Y=0 TO 8 STEP.25
80 LINE-(20*(Y+X),(Y-X+2+FNA(A))*6)
90 NEXT Y:NEXT X:GOTO 100
100 GOTO 100
```

NEW SOFTWARE - 40 GAMES

from Colin Chin

With 40 new games available from SVI, their release was an eagerly awaited event. Thanks to Microstyle Computers Ltd of Petone, I am pleased to quickly review them.

Generally, it was a mixture of very good, children's educational, good, and some wasted value games which I couldn't understand why SVI decided to release at all.

The best ones were ones that I bought, anyway. Some were almost in the category of Sasa, Ninja, but as the majority came from a Swedish stable, some also came from Israel, France and the USA.

JUMPING JACK was a really addictive game. I got into the 240s, then 400's, then managed to break the 600 and 800 barrier, occasionally. Many hours later, I hit the magic thousand, and surprise, surprise, I got a bonus life! I finally threw the towel at 1,240. But for value, it was one of the best.

Basically, it was a bouncy ball-type game, with the cursor acting as a monkey that had to bounce off the island with two coconut trees. BUT, everytime you bounced, you opened up a hole to the sea - so, if you landed on the same spot, you landed in the drink and lost out to the alligator lurking. On the upwards bounce, you had to collect all the coconuts on the trees EXCEPT the bad one, which promptly lost you another life. You did, however, start with three lives. Oh, if you find a snake walking along the ground, you could land on him, and grab another 50 points, but invariably, he was only around for a few seconds. The Joy Stick or cursor pad gave you control over the direction. Best of luck!

MAZE CUP CHAMPION

This was a tricky one, but had my youngest daughter really scratching her head wondering how I managed to beat the computer. Simply, you found yourself at the start of a big maze, trying to get to the Cup, and then escape from the maze with the trophy.

You had to get to the Cup first, otherwise you couldn't find your way out. AND you couldn't study the map too long because there was a built-in time-clock EVEN if you pressed the STOP key.

I did manage to find out how to win against the computer, and the graphics were quite imaginative, but when you hit the wall it went "Ouch". The timeclock worked against you in the higher levels, and you couldn't get another maze until you got through the previous one.

CRUNCH

Now here was a kong-type game that really got me going. And it was a change to find one, at last, available for the SV.

The graphics were a little uninspiring, but the action more than made up for this shortcoming. The ladders were effective, but the little dragons were a bit hammy but you still had to avoid them if you wanted to survive.

21

If you like card games - particularly "Pontoon", then this game is for you. It was an all action game which just kept going until you ran out of money.

The graphics were good, and the continuous action was marvellous. A really good recommendation for that quiet evening for card games.

SPECTRAFROG

Here at last - SV's answer to "Frogger". It wasn't a bad attempt, but was disappointing to find the frog a bit of a giant as compared with the moving cars. And also I couldn't get him to land on the cars, either! A welcome addition to the stable, still.

EDUCATIONAL

For Junior, filling the gap of not providing for the educational function has always been the stick being pointed at SVI. These may be part of the solution. At least it's a good start. The American ones are inspiring, and of a standard expected.

For the record:

CAKE BANDIT - maths

PRINCE & THE DRAGON - maths

*HARE & TORTOISE - Counting

*BUSY BEE - an artist type

*FIRST STEP -

*MUNCH-A-MATH -

* all rated A1

CRYPTIC CUBE, BONE OF CONTENTION, MINI GOLF, HORSE RACE and TROUBLE TROLLEY would be rated as Good in graphics, sound, playability and challenge. The Mini Golf has an inspiring drive routine, which I haven't managed to find for programming.

BOBO and KILLER CAR were difficult to play.

LIBRARY SOFTWARE EX AUSTRALIA

Members will have seen in the A'sian SVUG Newsletter, a list of Software available for sale. For those(out of town) who are unaware of what is available; here is a full list:-

METEOR STORM	Shooting	\$10.00
SPACE PILOT	Shooting	10.00
3D MAZE	Graphic Maze	5.00
MORMAR	Adventure	5.00
PACMAN	Chomp Chomp	10.00
ELIZA	Word Response	7.00
RESCUE	Shooting	10.00
SKYBASE	Shooting	10.00
HOUSE OF FRANKENSTEIN	Adventure	5.00
MARVN	see APC	2.00
MYSTERIOUS MANOR	27k Adventure	5.00
RURIKS' CUBE		5.00
FILING CABINET	22k Business	5.00

CRUNCH	BasicUtility	10.00
DISASSEMBLER	Utility	2.00
UNERA.COM	C/PM Utility	10.00
DISK.COM	C/PM Utility	10.00
ADHEX.COM	C/PM Utility	10.00
MFT.COM	C/PM Utility	2.00
XDIR.COM	CPM Sorted Dir	2.00
ZCPR.ASM(Full disc)	CPM Better CCP	2.00
ADVENT.COM (FD)	CPM Adventure	2.00
ALGOLM.COM (FD)	CPM Algol lang	2.00

PLEASE NOTE: The above prices are in \$A and are only available from the Librarian of SVAUG, Mr Jim Collins. In addition, there is a charge of \$A3.00 for cassettes OR \$A6.00 for each disk IF THESE ARE NOT SUPPLIED WITH ORDER. DON STANLEY has written to clarify the situation of supply of software to members of the S.V.U.G. (Wellington), as it may well be easier to arrange a bulk order. In addition, members if ordered individually, could well incur large amounts of Custom Duty/Sales Tax. Peter Deckert of the SVAUG has written saying he now has a modem, so if anyone has one - the Don'll be only too pleased to arrange downline software on behalf of all of us!

Another important note is that from the above prices, small royalties have been agreed to be paid back to the original authors- and thus protect by copyright, the originators. So for the small price asked, please respect our fellow SV users over the Tasman.

SV BOOKS AVAILABLE NOW

Now suddenly, there appears to be a multitude of publications available on the Spectravideo 318/328. CDL have promised an English translation of two Swedish books to be made available to dealers prior to Christmas.

To summarise the current position, the following are now available :-

SPECTRAVIDEO COMPUTING (Ian Sinclair - published by Granada Technical Books, U.K.) Price at \$24.95 from any SV dealer, and most bookshops. Any difficulty, please contact the Club Treasurer. Some shops have sold out, but the N.Z. agents have promised further supplies in November.

BASIC REFERENCE MANUAL FOR SVI318/328 PERSONAL COMPUTER SYSTEM (Written and published by SVI). Price \$24.95 and available from any SV dealer. A 366 page reference manual to overcome the shortcomings in the original 318/328 manuals. In addition, disc commands are also included.

SPECTRAVIDEO TECHNICAL & REFERENCE MANUAL (Compiled & Published by CDL, Auckland). Price approx \$20.00 from any SV dealer.

SPECTRAVIDEO ROM BASIC EXPLAINED (Written & published by the Spectravideo Australasian User Group at the price of \$A20.00). For further information, contact the Club Treasurer as a bulk purchase is being considered.

THE MISSILE & OTHER PROGRAMS for SPECTRAVIDEO 318/328 (Bernard Scott - published and available only from Aspect Software, Creswick, Vic.) at the price of \$A9.95 which includes p/pkg. This is a book (size A4) containing 20 programs of games, graphic displays, music & sounds library, novelties and educational listings. In addition, there are copious notes accompanying each program. A sample program called "FREEFALL" is in the club's hands and available from Dennis Timmons, Club Librarian for cost of photocopying at 2x5c + post. For further information, contact the Club Treasurer, as a bulk purchase is being considered.

SPECTRAVIDEO BASIC REWRITTEN (written & published by the Spectravideo-MSX User Group (Wellington) for \$15.00 + \$1.50 p/pkg.

PEEKs & POKEs IN SPECTRAVIDEO (written by Dean Whitmore & Published by Dean Whitmore on behalf of the New Plymouth Spectravideo User Group). Expected price to be about \$14.00.

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